

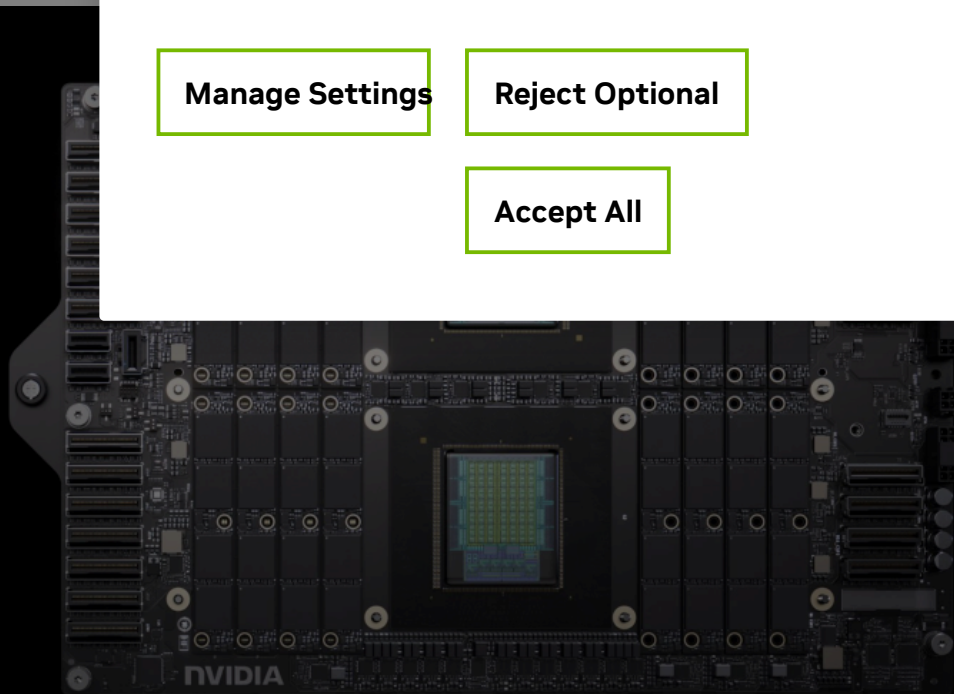
Data Science

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NVIDIA Perform Efficient

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Mar 16, 2026

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By [Praveen Menon](#) and [Ivan Goldwasser](#)



AI-Generated Summary



token production and improve productivity for model creators and users.

Modern GPUs operate at system performance a classic example of

This dynamic is especially training models with which enable AI agents software to complete

Both workloads commonly require strong single workstation. At the same time, concurrently, creating

The NVIDIA Vera CPU

- **Extreme single-core performance:** AI factories must sustain under constant load with many concurrent users and agentic tasks.
- **High memory and fabric bandwidth per core:** To ensure consistent SLA under load that moves volumes of data efficiently for real-time analysis and context switching tasks.
- **Efficient rack-scale co-design:** AI factories must rapidly deploy and manage capacity to fulfill agentic demand while maximizing power efficiency.

Data centers built with Vera maximize AI infrastructure investments, whether Vera CPUs are directly connected to accelerators or performing tasks on standalone CPU capacity at the end of a wire.

The post-training reality

Reinforcement learning requires models to constantly evaluate their outputs, recognizing which results succeed or fail. For example, models learning to do software development generate large amounts of code using models running on accelerators, which is then shipped to clusters of CPUs to build, run, and test—acting in a feedback-reward loop (see Figure 1).

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environments, each with a full complement of tools. Often, a single CPU core executes each lightly threaded case end-to-end from a set of accelerator-generated requests.

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Figure 1.

To maximize accelerator utilization, the generation and training phases of the model must be tightly coupled. If evaluation jobs running on a CPU finish jobs too late to influence the next step in the cycle. When this happens, it takes the model longer to learn to the same quality, and valuable tokens are wasted.

Agentic loops demand a unique blend of high single-core performance, massive data bandwidth, and deterministic execution with minimal tail latencies from the CPUs they employ.

These requirements are a central focus of the NVIDIA [Vera CPU design](#) (Figure 2), which delivers up to 50% faster sandbox performance compared to competitive platforms, 1.2 TB/s of memory bandwidth, and 88 Olympus cores with NVIDIA Spatial Multithreading (SMT) for the task concurrency necessary for AI Factories.

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NVIDIA Olympus

The need for higher-performance cores that support AI led to the NVIDIA Olympus core, the first fully custom data center CPU core from NVIDIA. Olympus debuts in Vera alongside the second generation of the NVIDIA Scalable Coherency Fabric (SCF), originally developed for the NVIDIA Grace CPU.

Built for sustained high Instruction Per Cycle (IPC) operation on memory-intensive workloads with control-flow logic, Olympus uses a 10-wide instruction fetch and decode frontend, and a neural branch predictor capable of evaluating two taken branches per cycle. It is fully compatible with the Arm v9.2 instruction set and existing software for high performance on Arm-based containers, binaries, libraries, and operating systems.

Users can choose between performance-per-thread and thread count at runtime with NVIDIA SMT. This gives each thread stable performance, stronger isolation, and predictable tail latency under heavy load. Traditional SMT relies on time-shared resources and frequent context switching between threads, introducing performance variation.

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The Vera CPU is built implementing memory topology.

From the point of view like other cores, each throughput bandwidth cross-die traffic typi

The runtime paths of control planes are in topology of the proc considered ahead of performance without

The second-generation subsystem, delivering

Each core is provisioned with up to 14 GB/s of memory bandwidth, roughly 3x the per-core rate of traditional data center CPUs—ensuring Extract-Transform-Load (ETL), real-time analytics, and memory-bound workloads maintain throughput when every core is active.

Feeding SCF is Vera's second-generation LPDDR5X memory subsystem, delivering up to 1.2 TB/s of total bandwidth at less than half the memory power of traditional DDR configurations and up to 1.5 TB of capacity—a 3x increase over the prior generation. Small Outline Compression-Attached Memory Modules (SOCAMM) brings low-power memory into the data center for the first time, replacing soldered memory with detachable, upgradable modules that combine LPDDR efficiency with server-class serviceability.

Performance across the AI factory

All these architectural elements enable the Vera CPU to deliver up to 1.5x the agentic sandbox performance under full-socket load compared to competitive x86 platforms across compilers, scripting tools, runtime engines, compression, and agentic tool calls (Figure 3).

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Figure 3. NVIDIA Vera offers up to 1.5x higher sandbox performance compared to the competition

This advantage compounds across three dimensions. In RL post-training, a 1.5x faster sandbox returns evaluation results within tighter time windows, enabling models to capture the best gradient tokens and accelerating training cycles.

In agentic inference, it reduces users' wait time, improving accelerator utilization and easing pressure on KV cache offloading.

For frontier training problems, 50% higher single-core performance means more sequential tests complete before hitting time limits, expanding the range of hard problems a model can learn from.

Agentic environments by the rack

Every AI Factory requires millions of CPU cores to enable the agentic loop of RL and tool use. To unlock the potential of AI infrastructure, deployment must be rapid. For many AI factory

The new NVIDIA Vera CPU Rack offers incredible density and performance within the same planning constraints as traditional racks, while being deployed today.

With a capacity of more than 2x the performance of traditional racks, manage capacity at a lower cost and market for new capacity.

Each Vera CPU is controlled by a centralized management system, ensuring the most efficient use of resources.

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Vera platforms and configurations

In addition to the Vera platforms for the diverse densities, cooling capabilities, and power constraints, our partners are enabling space available in an

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Platform		
NVIDIA Vera Rubin NVL72		
NVIDIA Vera CPU Rack		
Single and dual-socket Vera platforms	Flexible server platforms built around one or two Vera CPUs, with up to 1.5TB LPDDR5X per socket and 1.8TB/s NVLink-C2C between CPUs in dual-socket designs, suitable for any facility.	Cloud infrastructure, enterprise, analytics, storage, HPC, NVIDIA PCIe GPU-equipped servers, and AI factories.
NVIDIA HGX Rubin NVL8	Accelerated computing platform pairing Vera host CPUs with Rubin GPUs over PCIe, enabling balanced CPU-GPU performance across multiple server designs.	AI inference, technical computing, analytics, and enterprise HPC deployments.

Table 1. Vera platform options for modern AI factories

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Platform availability

Vera systems will be available from major OEMs, including Cisco, Dell, HPE, Lenovo, and Supermicro, in the second half of 2026. See the [Vera CPU webpage](#) for more details.

Learn More about the [Vera CPU](#) and [Vera Rubin](#).

NVIDIA Vera performance compared to AMD EPYC Turin and Intel Xeon 6 Granite Rapids, across a variety of workloads, including code compilation, interpreters, scripting, runtime engines, ETL, data analytics, and graph.

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